Cover letter

Mathematics of Computation 25 October 2026

Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript "A certified sixteen-prime-factor lower bound for Lehmer's totient problem" by Deep Bhattacharjee.

The existing sixteen-factor lower bound and companion results are retained. A new self-contained appendix proves termination and completeness of a finite construction for every prescribed factor count k. It does not prove that all those outputs are empty; the full Lehmer conjecture remains open.

The finite reductions, reproducible computational number theory, independently checked searches and primality certificates make a computational-number-theory journal the clearest fit.

The prior public preprint is `https://hexagonmath.org/2610.00454v1`. The research manuscript/source archive is `https://doi.org/10.5281/zenodo.23629631`. Editable sources and verification records are supplied with the manuscript.

The repository contains partial Lean results for statements not depending on the exhaustive search. The full searches and the full Lehmer conjecture are not claimed to be Lean-formalised. The new uniform construction has a written proof and fresh exact controls; it is not Lean-formalized. The guarded k=14 run is explicitly incomplete.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The earlier archived manuscript had a human-understanding confirmation. Added research material awaits author review. The named human authors retain responsibility for the content.

Sincerely,

Deep Bhattacharjee
Corresponding author
itsdeep@live.com
ORCID: 0000-0003-0466-750X